

L12b: Why Do Markets Crash?

CHEME 5660 Cornell University

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Risk. Trading involves risk and is generally not recommended for those with limited resources, experience, or a low risk tolerance. Past performance does not guarantee future success.

Responsibility. You are responsible for your investment decisions based on your financial circumstances, objectives, risk tolerance, and liquidity needs.

Objectives for Today

Today, we build a simple agent-based market and ask whether changing one entry of a trading rule can turn a calm market into a falling one.

Today's concepts

- Rules as policies: a lookup table from what neighbors intend to what a trader intends next.
- The dealer: how net buying raises the price, net selling lowers it, and why a switched market needs a control.
- Rule-switch simulation: reading the fraction of Sell intentions and the price, and what the comparison does and does not show.

What You Need to Know

Core concepts

- A Wolfram rule maps the neighbor sum $c \in \{0, 1, \ldots, 8\}$ to the trader's next intention. Some traders switch from a baseline rule R to a switch rule $\bar{R}$.
- The dealer sets the price from the net trade. We compare every switched market with a control that shares its history.
- After a fraction f of the traders switch, the Sell intention can spread from neighbor to neighbor, and the price falls while the sellers have shares. This is an analogy for crashes, not an explanation of them.

For understanding (you won't reproduce these)

- The dealer's price update and the two algorithms.
- The technical traders' band rule and their edge conditions.
- The Q-learning notebook.

Lectures and Examples

Lecture:
CHEME-5660-L12b-Lecture-WhyDoMarketsCrash-ABM-Fall-2026.ipynb

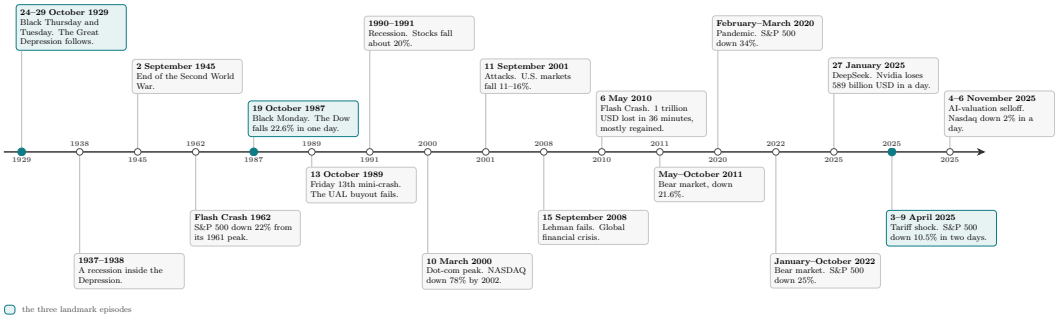
Examples:
CHEME-5660-L12b-Example-Wolfram-Rules-Fall-2026.ipynb
CHEME-5660-L12b-Example-Wolfram-RuleSwitch-Fall-2026.ipynb

Review on your own:
CHEME-5660-L12b-Example-Wolfram-StylizedFacts-Fall-2026.ipynb
CHEME-5660-L12b-Example-Wolfram-Viewport-Fall-2026.ipynb

Optional advanced notebooks are listed on the Optional Advanced Material frame.

Why Do Markets Crash?

A market crash is a sudden, significant decline in prices across a broad market. The three teal episodes are described on the next frame:



Three Landmark Episodes

Three episodes, each with a different trigger:

- 1929 Wall Street Crash: investors had bought shares with borrowed money (on margin). When prices fell, brokers demanded more cash, and many had to sell.
- 1987 Black Monday: portfolio insurance, often done by selling stock-index futures as prices fell, added selling to a falling market. The Dow fell 22.6% in one day.
- April 2025 tariff mini-crash: tariff announcements and fears of retaliation brought sharp declines and a jump in volatility. Stocks recovered much of the loss after the higher tariff rates were paused for most countries on April 9.

Sources: Federal Reserve History; Brady Report; USTR; Federal Reserve FSR

Agent-Based Markets

Markets are man-made systems, and they can fail. An agent-based model simulates traders as autonomous agents with their own rules, and the price emerges from their actions:

- Iori (2002): interacting traders who imitate each other produce large fluctuations and volatility clustering.
- Lux–Marchesi (1999): agents switch between fundamentalist and chartist strategies, producing bubbles and crashes.

Today, traders sit on a grid and act on what their neighbors intend to do.

Sources: Iori; Lux and Marchesi, Nature 397

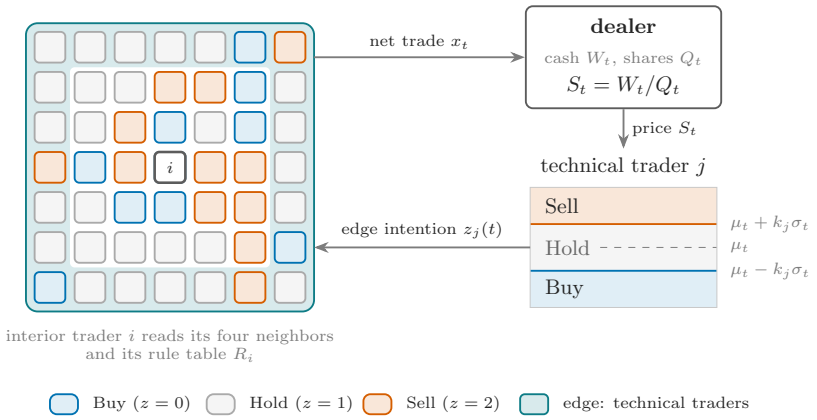
Agents as Policies

A policy is a table that gives, for each possible state of an agent's surroundings, the action the agent takes. In our market:

- State: a number summarizing what a trader's four neighbors intend.
- Action: the trader's own next intention, Buy, Hold, or Sell.
- Source of the table: Q-learning could learn it from experience. Our traders are handed their tables, and we change one entry.

Wolfram Markets

Traders on a grid send orders to a dealer, and the dealer's price returns to the grid through the technical traders on the edge:



Wolfram Markets: The Grid and Intentions

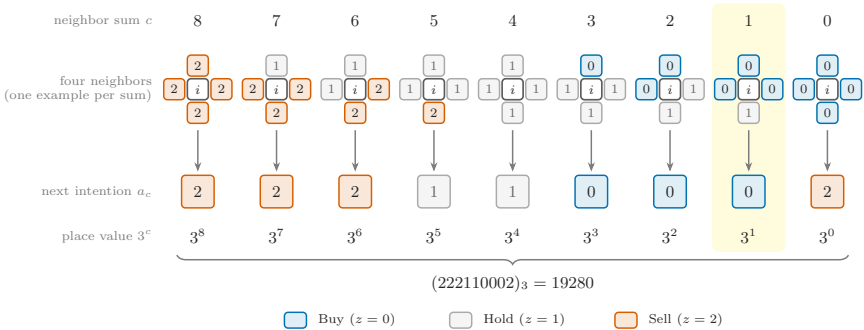
Traders sit on the cells of a square grid, and simple local rules produce market-level behavior, as in Wolfram's cellular automata, grids in which each cell updates from its neighbors by a fixed rule:

- Intention: Buy, Hold, or Sell, coded 0, 1, 2. We write $z_j(t) \in \{0, 1, 2\}$ for the intention of trader j at step t.
- Interior traders: each has four neighbors, above, below, left, and right, and updates its intention from theirs.
- Edge: some edge traders are technical traders who react to the dealer's price. The other edge positions stay in Hold.

Source: our paper on rule switches in Wolfram markets; Wolfram, *A New Kind of Science* (2002), Chapter 3

Traders and Their Rules

Trader i sums the intentions of its four neighbors ∂i, $c_i(t) = \sum_{j \in \partial i} z_j$, and reads its next intention $z_i(t) = a^{(R_i)}_{c_i(t)}$ from its rule R_i. The nine entries a_c, read in base 3, give the rule number $\boxed{R = \sum_{c=0}^{8} a_c \, 3^c}$. Rule 19280 is one example:



The Dealer

- Price: the dealer quotes $S_t = W_t/Q_t$ (cash over shares) and keeps $W_t Q_t = W_{t-1} Q_{t-1}$ (constant product). After the net trade x_t, $Q_t = Q_{t-1} - x_t$, and every share trades at $F_t = W_{t-1}/Q_t = \sqrt{S_{t-1} S_t}$. Net buying ($x_t > 0$) raises the price.

- Depth: the dealer starts with $Q_0 = hN_{\text{tr}}$ shares for N_{tr} traders, and a larger depth h gives a smaller move.

- Price noise: a course addition that scales the dealer's cash just before the trade. Each step lasts Δt years, the draw is $Z_t \sim \mathcal{N}(0,1)$, and σ_S is the annualized volatility of the price noise:

$$\boxed{S_t = S_{t-1}\left(\frac{Q_{t-1}}{Q_{t-1} - x_t}\right)^2 \exp\left(\sigma_S \sqrt{\Delta t}\, Z_t\right)}$$

The Dealer and the Technical Traders

- Growth rate: the growth rate over step t follows from the price update:

$$g_t = \frac{1}{\Delta t} \ln \frac{S_t}{S_{t-1}} = \underbrace{\frac{2}{\Delta t} \ln \frac{Q_{t-1}}{Q_{t-1} - x_t}}_{\text{traders}} + \underbrace{\frac{\sigma_S}{\sqrt{\Delta t}} Z_t}_{\text{price noise}}$$

  A small net trade gives the Farmer–Joshi form, with half of the dealer's shares as the liquidity $\lambda_t = Q_{t-1}/2$: $g_t \approx x_t/(\lambda_t \, \Delta t) + (\sigma_S/\sqrt{\Delta t}) \, Z_t$.

- Technical traders: track a moving average μ_t and a moving departure σ_t of the price, with smoothing $0 < \alpha < 1$. Trader j intends Buy below its band $\mu_t \pm k_j \sigma_t$, Sell above it, and Hold inside it:

$$\boxed{\mu_t = \mu_{t-1} + \alpha \, (S_t - \mu_{t-1}) \qquad \sigma_t = \sigma_{t-1} + \alpha \, (|S_t - \mu_t| - \sigma_{t-1})}$$

Algorithm: One Market Step

Each step $t = 1, 2, \ldots, T$ runs in this order, with switch step t_s. Interior trader i follows rule R_i, which starts at the baseline rule R:

1. Switch: at $t = t_s$, the chosen fraction f of interior traders get the switch rule $\bar{R}$ and keep their intentions, cash, and shares.
2. Clear: the dealer's cash gets the price noise. The dealer then clears the orders from $z_i(t-1)$ and returns x_t, Q_t, W_t, and S_t.
3. Band: update μ_t and σ_t from S_t.
4. Edge: technical traders set $z_j(t)$ from the band (active), Hold (held), or a random draw (random signal).
5. Interior: every interior trader computes $c_i(t)$ from the old interior and new edge intentions, and sets $z_i(t) \leftarrow a_{c_i(t)}^{(R_i)}$.

Full version: the lecture notebook, after Algorithm S1 of our paper

Algorithm: Dealer Clearing

The dealer finds one price $F(x) = W_{t-1}/(Q_{t-1} - x)$ for a net trade x, with excess demand $e(x)$ of buyers who can pay minus sellers with a share:

1. Direction: $d \leftarrow \operatorname{sign} e(0)$, and bounds ℓ, u.
2. Search: bisect for the largest m with $d \cdot e(dm) \geq m$.
3. Price: $x_t \leftarrow d\ell$ and $F_t \leftarrow F(x_t)$.
4. Fill and settle: fill the short side in full; ration the other side by a seeded priority; move cash and shares at F_t.
5. Dealer: $Q_t \leftarrow Q_{t-1} - x_t$, $W_t \leftarrow W_{t-1}Q_{t-1}/Q_t$, and $S_t \leftarrow W_t/Q_t$.

Full version: the lecture notebook, after Algorithm S2 of our paper

Example: One Rule

Example:
CHEME-5660-L12b-Example-Wolfram-Rules-Fall-2026.ipynb

In this example, we read the table of one rule and run a market in which every interior trader follows it. We then run a second market under a different rule, and compare the two grids and the two share prices.

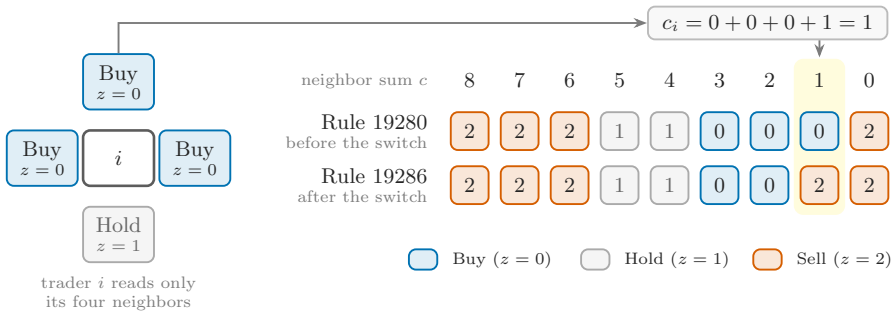
Example: Does the Price Look Real?

Review on your own:
CHEME-5660-L12b-Example-Wolfram-StylizedFacts-Fall-2026.ipynb

In this example, we measure the stylized facts of the dealer's price in a market that never switches, compare them with a real ticker and the Farmer–Joshi market, and vary the edge traders and the dealer's depth.

The Rule Switch

A rule switch moves some traders from a baseline rule R to a switch rule $\bar{R}$. The algorithm accepts any two different rules, and we study pairs that differ in one entry. Here, with three Buy neighbors and one Hold ($c = 1$), Rule 19280 buys and Rule 19286, $(222110022)_3$, sells:



The Rule-Switch Experiment

We use the one-entry difference as a model of a change in psychology. The experiment has three parts:

- Shared history: all interior traders follow the baseline rule R. At the switch step, a fraction f of them, chosen in advance, receive the switch rule $\bar{R}$ and keep their intentions, cash, and shares.
- Control: a second market with the same history never switches. Differences after the switch come from the rule change.
- Edge traders: active, held, or random signal (a course addition). When held, any cascade comes from traders reading their neighbors.

Example: Rule Switch

Example:
CHEME-5660-L12b-Example-Wolfram-RuleSwitch-Fall-2026.ipynb

In this example, we run a switched market and its control, and watch the Sell intention spread across the grid. We then follow the share price as it falls and levels off, and bring in fresh players to move the share price again.

Example: Closed and Open Markets

Review on your own:
CHEME-5660-L12b-Example-Wolfram-Viewport-Fall-2026.ipynb

In this example, we run a rule switch in a closed market, in which the same traders stay on the grid, and in an open market, in which traders leave and new players take their places. We then let a fraction of the players follow a contrarian rule, and compare the share price and the filled orders with and without the contrarian players.

Optional Advanced Material

Q-learning: advanced/qlearning/
CHEME-5660-L12b-Advanced-QLearning-Fall-2026.ipynb

How could a trader learn a policy table from experience instead of being handed one? We write the Q-learning update, lay out the algorithm, and describe when it converges to the best policy.

Summary

We built a simple agent-based market and used it to ask how a small change in trader behavior can tip a market toward selling.

Key takeaways

- Rules as policies: a rule is a lookup table, and one entry can change.
- Dealer and control: net trades set the price, and a control isolates the switch.
- Rule switch, by analogy: the Sell intention spreads, and the price falls until the sellers run out of shares. A simulation does not show that a rule change causes real crashes.

Agent-based models ask how individual rules add up to market behavior.